Universidad de los Andes
Facultad de Economía

Big Data and Machine Learning para Economía Aplicada
MECA 4107
Créditos: 4
2026-20

Problem Set 2: Predicting Poverty

"Wars of nations are fought to change maps.

But wars of poverty are fought to map change" M. Ali

1 Introduction

This problem set was inspired by a recent competition hosted by the World Bank: Poverty Prediction Challenge. The idea is to predict poverty in Colombia. Accurate poverty measurement is essential for directing development efforts and informing evidence-based policies, yet many countries lack recent data due to the high costs and complexity of collecting comprehensive household expenditure surveys. By building better predictive models, policymakers can design shorter, cheaper surveys that rapidly identify poor households and target interventions more effectively.

The objective is to predict poverty at the household level. Data are provided at both the household and individual levels, and you can use individual level information to build additional variables that improve your prediction.

An essential dimension for policymakers is that they can *rapidly and cheaply* measure poverty. A model that predicts well but relies on variables that are expensive to collect or hard to verify may be of limited practical value. This tension between predictive performance and policy usefulness is at the heart of this problem set:

What is the best model to predict poverty, and what would it imply for public policy?

This question should inform your choices from the very first stage.

2 General Guidelines

2.1 Deliverables

Each team must submit the following materials:

- **Slides for in-class presentation.** Each team must prepare **two** separate slide decks. Each slide deck should be submitted as a `.pdf` file and uploaded to the corresponding activity in Bloque Neón before class. Files must follow the naming convention: `nombre_equipo_XX.pdf`, where `nombre` is `best` or `compare`, and `XX` is the team number (use a leading zero for team numbers below 10). For example, for Team 1 the files would be:
  - `best_equipo_01` (Best Model Deep Dive)
  - `compare_equipo_01` (Algorithm Comparison)

- **Public GitHub repository.** Each team must submit a link on Bloque Neón to a **public** GitHub repository containing all code required to reproduce the analysis and predictions. The repository must follow the provided template and show at least five (5) substantial contributions from each team member on the main branch. Code must be fully reproducible, readable, and well-commented. I encourage you to follow the tidyverse style guide. The README should help the reader navigate your repository; for example, Project Awesome has a curated list of interesting READMEs.

- **Kaggle submissions.** Predictions must be submitted on Kaggle. To join the competition use the following link. Each team must submit at least **fifteen (15) predictions** using models trained with at least **five (5) distinct classification algorithms**[1] chosen from the following: *Linear Probability Model, Logistic Regression (Logit), Elastic Net, CARTs, Random Forest, Boosting, and Naive Bayes.*

2.2 Presentation Format

Each team will deliver a **15-minute in-class presentation**. On the day of the presentation, each team will be randomly assigned which of the two decks to present. All team members must be equally prepared to present either deck.

Presentations should clearly communicate the empirical analysis to an audience of economists. Each slide deck should tell a coherent empirical story. Focus on interpretation and conclusions rather than mechanical reporting. Tables and figures should be self-contained, properly formatted, and publication-quality, with a title, labeled axes, and a legend. Tables must not be screenshots from R, Python, or any other software. No code in the slides. Avoid excessive text.

Golden Rule: Every slide should earn its place: if it does not advance the story the deck is telling, remove it.

2.3 Evaluation Criteria

A full rubric for the presentation and repository can be found in Bloque Neón.

[1] Kaggle imposes a daily limit on the number of submissions. Plan accordingly and do not leave submissions for the last day.

3 Building the Predictive Model

To make this concrete, note that a household is classified as

$$Poor = I(Inc < Pl) \tag{1}$$

where I is an indicator function that takes one if the family income is below a certain poverty line. This suggests two routes: predict poverty directly as a classification problem (zeros and ones), or predict income and then apply the poverty line. In this problem set you will take the classification route.

Building a predictive model is an iterative process. You will train models, submit predictions to Kaggle, receive feedback from the public leaderboard, and use that feedback to refine your approach: adjusting features, tuning hyperparameters, or trying new algorithms. Each iteration should be reflected as a meaningful contribution in your GitHub repository. The stages below describe the components of this process; the slide decks present the final result of this iterative process.

The predictive modeling process can be organized into four stages.

3.1 Stage 1: Understanding the Data

The data comes from DANE and the mission for the "Empalme de las Series de Empleo, Pobreza y Desigualdad - MESE". The data contains four sets divided into training and testing at the household and individual levels. You can use the variable `id` to merge households with individuals. You will note that some variables are missing in the testing data sets; this is designed to make things a bit more challenging. More information about the data is available at the competition website.

With this data in hand, the first task is to understand it. This means cleaning and merging the household and individual level datasets; constructing new variables from individual level information that may improve household level prediction; and examining the structure of the outcome variable: its distribution, the degree of class imbalance, and its relationship with potential predictors. Use your knowledge as social scientists to guide your exploration. What variables should predict poverty based on economic reasoning? What patterns in the data confirm or challenge those priors? How severe is the class imbalance, and what does that imply for model training?

3.2 Stage 2: Training and Comparing Algorithms

With a clean dataset and a set of candidate features, begin training classification models using the required algorithms listed in the deliverables. Start simple, submit to Kaggle, and use the leaderboard feedback to iterate: try different feature sets, adjust hyperparameters, and refine your approach. For each algorithm, select hyperparameters through a systematic process: define the range of values to explore, choose a search strategy (e.g., grid search, random search), and use cross-validation to guide your choices. Once you have the best submission from each algorithm,

compare them: go beyond Kaggle scores and try to understand *why* some algorithms outperform others on this task. What role do specification choices, tuning, feature engineering, and class imbalance handling play in the performance differences?

3.3 Stage 3: The Best Model

From the models trained in Stage 2, identify the one that achieved the team's highest public score on Kaggle. Now ask yourself: why did this model win? What is the algorithm doing that the others are not? Examine the hyperparameter configuration that led to the best performance. Investigate feature importance: which variables are the strongest predictors and why? What is the model capturing about the correlates of poverty? Consider whether a class imbalance strategy made a difference, and if so, why.

3.4 Stage 4: From Prediction to Policy

Your submissions are evaluated using the F1 score. Think carefully about what this metric means in a poverty targeting context. The F1 score balances precision and recall, but each type of error has a different policy cost. False negatives are poor households that the model classifies as non-poor: these families are excluded from the program (undercoverage). False positives are non-poor households that the model classifies as poor: these families receive benefits they do not need (leakage). How does your best model balance these two types of errors? What does this trade-off imply for the design of a targeting instrument? Would a policymaker prefer a different balance than the one the F1 score rewards?

Beyond the model's errors, consider whether it could be deployed in practice. Could the best model be implemented with a short, cheap survey? Are the key predictors easy to collect and verify in the field, or would the model be difficult to operationalize? What would you recommend to a government agency designing a poverty targeting instrument?

4 Slide Decks

Each deck maps onto the stages above.

- `best_equipo_##` **Best Model Deep Dive.** This deck tells the story of your best model from the data up and connects it to policy. The first slide is mandatory (see below). Beyond that, teams are free to organize their deck, but a recommended structure is:
  - *Mandatory first slide:* A *Best Model Overview* slide with the specification of the best model and the Kaggle score (public leaderboard). State the main takeaways upfront and begin constructing the story.
  - *Data (2–3 slides):* Use your domain knowledge as social scientists to show the audience why the variables in your best model matter: what economic reasoning led you to consider them, and what the data revealed.

  - *Best model (4–5 slides):* Deep dive into the training process, hyperparameter tuning, feature importance, and diagnostics.
  - *Policy implications (1–2 slides):* Discuss what the F1 score reveals about your best model's errors: how does it balance leakage and undercoverage, and what does that imply for poverty targeting? Could the model be deployed in practice with a short, cheap survey, or do the key predictors make it difficult to operationalize?

- `compare_equipo_##` **Algorithm Comparison.** This deck positions the best model against the alternatives. The first slide is mandatory (see below). Beyond that, teams are free to organize their deck, but a recommended structure is:
  - *Mandatory first slide:* A *Best Model Overview* slide with the specification of the best model and the Kaggle score (public leaderboard). State the main takeaways upfront and begin constructing the story.
  - *Data (1–2 slides):* Use your understanding of the data and the problem to explain why certain algorithms and specifications may work better than others.
  - *Algorithm comparison (4–5 slides):* Present a summary of all required algorithms, then focus the detailed comparison on the two algorithms that came closest to your best model's performance. A meaningful comparison requires a level playing field: differences in scores should reflect the algorithm, not differences in the set of predictors or preprocessing choices. Use the deep dive into the top contenders to explain what drives the performance gap.
  - *Conclusion (1 slide):* What the comparison reveals about the trade-offs between model complexity, interpretability, and predictive performance. If a government agency asked you to recommend one model for poverty targeting, would it necessarily be the one with the highest Kaggle score?